

## What is Operator Overloading

Operator overloading is a feature in C# (and many other OOP languages) that allows you to redefine how operators (like +, -, \*, /, ==, etc.) work for your own custom types (classes or structs).

Instead of being limited to built-in types (int, double, etc.), you can make these operators behave meaningfully for objects you create.

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### Key Points

- Only classes and structs can overload operators.
  - You can't create new operators, only redefine existing ones.
  - The keyword operator is used in the method signature.
  - Operator methods must be declared public static.
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### Why Use Operator Overloading?

- Makes code more intuitive when working with objects.
  - Allows mathematical or comparison logic to be written naturally.
  - Improves readability when your object represents something mathematical (like complex numbers, vectors, fractions, etc.).
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Operator overloading lets you teach your class how to behave when someone writes `obj1 + obj2` or `obj1 == obj2`.